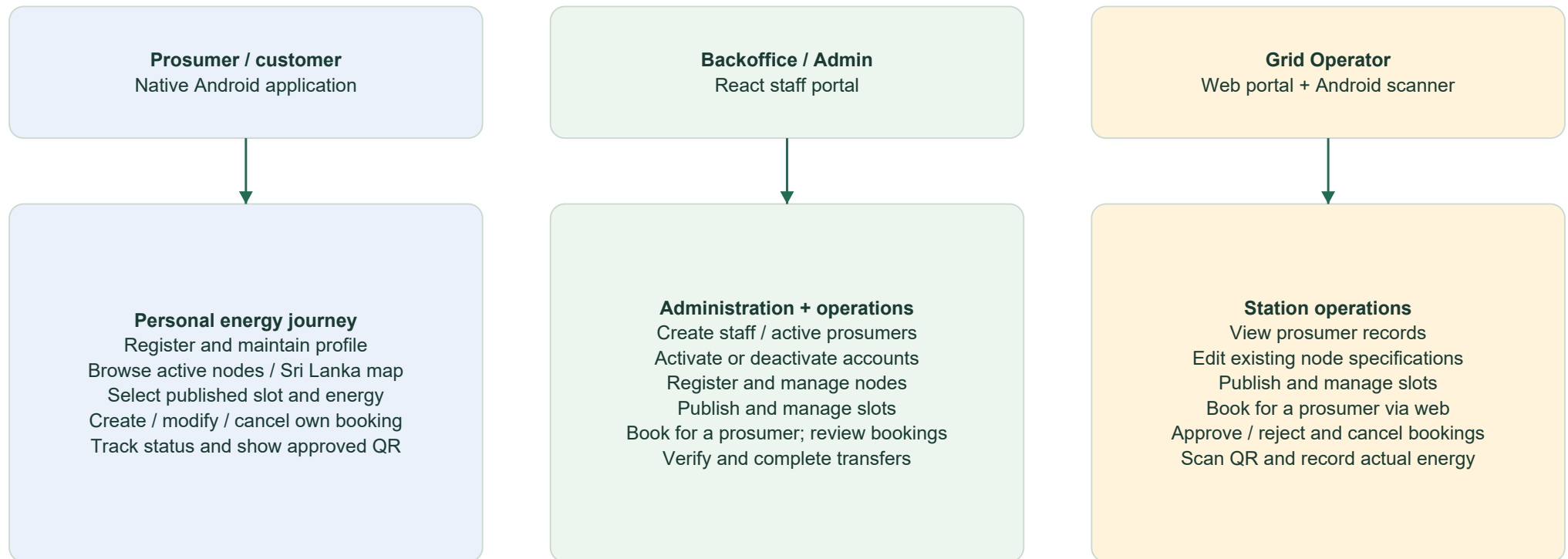


Roles and the complete workflow

01 / Plain-English guide to the current Solara implementation



Read in this order: role permissions, setup, booking journey, lifecycle, QR transfer, and practical examples.

User means Prosumer. Admin means Backoffice. Energy generation capacity uses kW; booked and transferred energy uses kWh. Physical battery/grid equipment is outside this software implementation.

Who can do what?

Permissions below describe the server API. The next page explains where each feature appears in the UI.

Function	Prosumer	Admin / Backoffice	Grid Operator
Register / profile / deactivate	Self-register; own profile; self-deactivate	Own profile; create staff and prosumers; activate/deactivate accounts	Own profile; view prosumer records
Station / node management	View active nodes and map	Create, edit, activate, deactivate, archive	Edit existing nodes; cannot create or activate
Trading slot management	View future slots	Publish, edit and archive unreserved slots	Publish, edit and archive unreserved slots
Create reservation	For self only	For a selected active prosumer	For a selected active prosumer
Modify / cancel reservation	Own Pending/Approved bookings; 12h notice	Any Pending/Approved booking; 12h notice	Any Pending/Approved booking; 12h notice
Approve / reject request	No	Pending bookings before start	Pending bookings before start
QR and transfer completion	Display own approved QR	Retrieve QR; verify; complete transfer	Retrieve QR; scan/verify; complete transfer
History / dashboard	Own reservations; active-node count	All reservations and operational counts	All reservations and operational counts

Which application does each role use?

A permission in the API does not imply that every client displays a button for it.

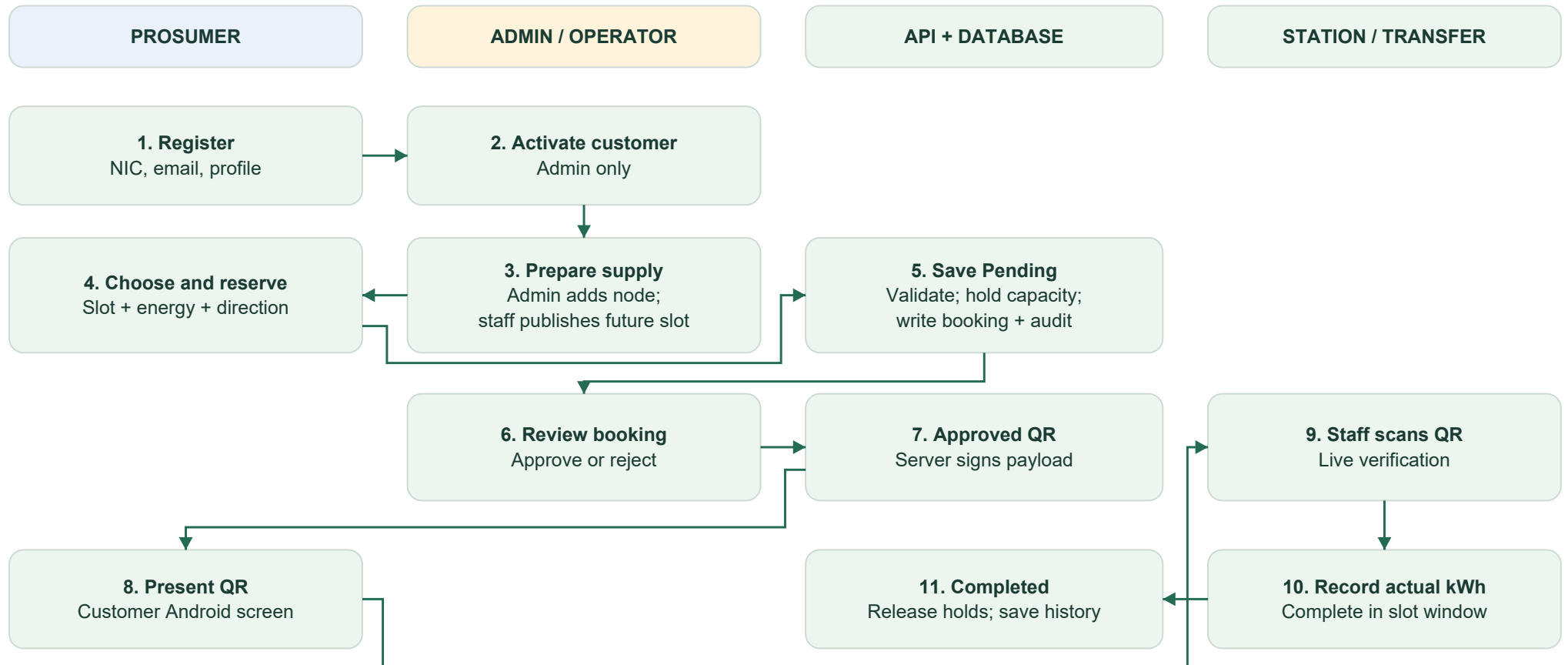
Client / role	Visible functionality	Important distinction
Prosumer Android	Register/login; overview; create/modify/cancel bookings; details/history; approved QR; map; profile; account deactivation; connection settings	Booking status applies immediately when selected. Text search has a Search button.
Admin web portal	Overview; reservations; node and slot management; user administration; profile; QR payload verification and completion	Prosumer login is directed to Android. Admin can create staff and active prosumer accounts.
Grid Operator web portal	Overview; create/review/edit/cancel reservations; edit nodes; manage slots; verify/complete transfers; own profile	Create reservation selects an active prosumer. No customer-confirmation step is implemented.
Staff Android	Operational dashboard/history; reservation details and review; QR camera scan/manual payload; complete transfer; map/profile	The current Android UI hides New reservation for staff. Staff create bookings through the web portal.

Customer consent and staff-assisted booking

The usual journey starts with a prosumer request. The current API also permits either staff role to create a Pending reservation on behalf of an active prosumer. It does not collect or enforce customer consent. For assisted phone/in-person requests, consent is an operating procedure rather than a software approval step.

End-to-end workflow diagram

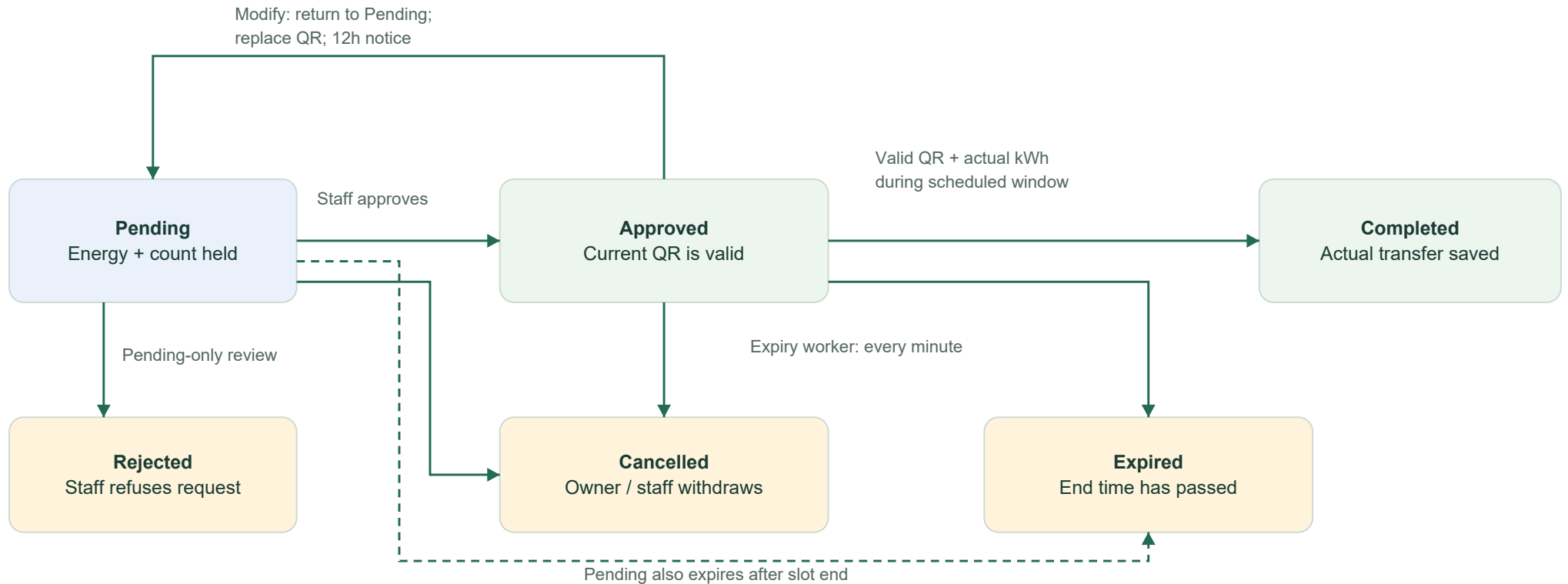
Main successful path; alternate outcomes are explained on the following page.



Admin account activation and station registration are setup actions. Grid Operators can publish slots and review bookings. Staff-assisted creation enters the same Pending state as a customer-created request.

Reservation lifecycle

Pending and Approved both hold energy capacity and one booking/battery allocation.



Action	Rule / effect
Modify	Owner or staff; old and new start must satisfy 12h notice. Revalidate new slot, release old hold, claim new hold, return to Pending and invalidate old QR.
Terminal outcomes	Rejected, Cancelled, Completed and Expired release the reservation hold. History remains. Terminal reservations cannot be modified or cancelled.

Cancel, reject and expiry

These actions have different meanings and time rules.

Action	Who / eligible state	When / result
Cancel	Prosumer owner or either staff role; Pending or Approved	At least 12h before start; status Cancelled; capacity released; QR invalidated.
Reject	Either staff role; Pending only	Before slot start; status Rejected; capacity released. This is the staff review decision.
Expire	Background worker; Pending or Approved	After slot end; runs every minute and handles up to 500 candidates per pass; status Expired; capacity released.
Complete	Either staff role; Approved only	Valid current QR, within start/end, positive actual energy no greater than reserved; status Completed.

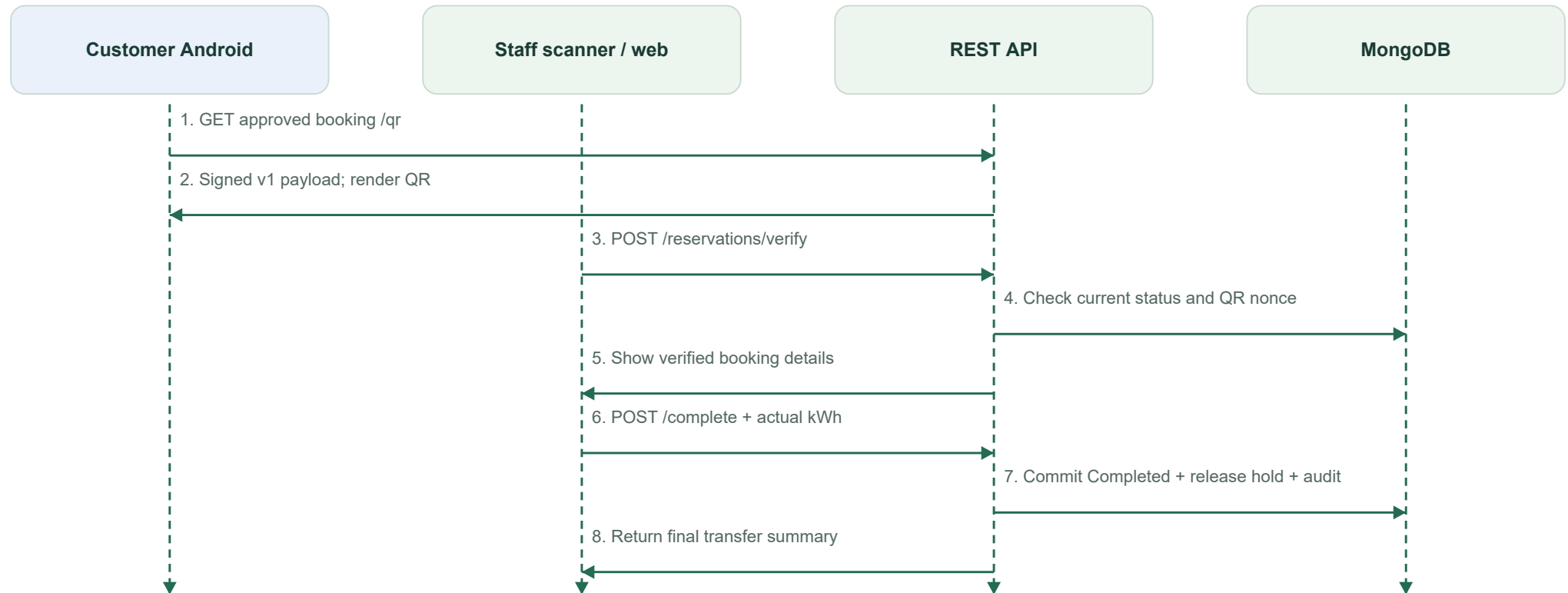
What date is the reservation for?

A booking inherits the selected trading slot's start and end. The prosumer does not choose a separate appointment date. A new reservation start must be in the future and no more than seven days from server time. Exactly seven days is accepted.

Example: selecting a slot on 12 October, 10:00-12:00 creates a booking for that window, even when the request is submitted on 10 October.

QR verification and energy transfer

The server creates a signed payload. Android draws the QR image; staff scan or enter its payload.



Verification is not completion. Verification can happen before the slot starts; completion must happen during the scheduled window. The current QR cannot complete a booking twice. For a 10 kWh reservation, 8 kWh may be recorded; 12 kWh is rejected.

Drop off, charging and the full-island map

Energy direction is a booking attribute; the physical transfer mechanism is not implemented.

Term / feature	Meaning in this project
DropOff	Energy from prosumer to microgrid/station. A compatible battery discharge or connected-grid export could implement this physically, but neither hardware model is specified in code.
Charging	Energy from microgrid/station to the prosumer battery/storage. It is not explicitly an EV-car charging system.
Energy measurements	Staff manually enter actual transferred kWh. No meter telemetry, inverter command, charger protocol, pricing or payment workflow is implemented.
Prosumer map	Loads GET /stations without radius parameters. Shows active registered nodes across Sri Lanka and opens with the island in view. Location permission only displays the user position.
Map freshness / limits	Refresh reloads station data; offline cache is labelled. Server station listing currently caps at 500 records. Demo/synthetic pins are not evidence of real stations.

Implementation references

These are repository paths, not external product documentation. The code is the source of truth for this guide.

backend/SolarTrading.Api/Controllers/ApiControllers.cs

backend/SolarTrading.Api/Services/ReservationService.cs and BusinessRules.cs

frontend/android/app/src/main/java/com/solara/microgrid/MainActivity.java and MapActivity.java

frontend/web/src/main.jsx